Surface Water Quality Index Estimation via Gradient Tree Boosting and Stoichiometric Feature Engineering

Semester 5 Applied Machine Learning (AML) | Project 18: Water Quality Prediction

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***Abstract*—Automated surface water quality monitoring is vital for public health safeguarding, ecological conservation, and municipal supply management. Traditional laboratory assays, such as 5-day Biochemical Oxygen Demand (BOD5), suffer from a prohibitive 120-hour incubation latency. This paper presents Phase 1 of an applied machine learning investigation predicting the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI) across 2,827,977 multi-national sensor observations spanning 83 years (1940–2023) across 5 nations (England, USA, Ireland, China, and Canada). A data quality audit disproved legacy claims of missingness, confirming 100% complete-case ground truth. Thermodynamic boundary screening purged 8,360 corrupted telemetry entries (0.296%), yielding 2,819,617 pristine observations. We engineered 38 domain features incorporating non-linear log(1+x) transformations, aquatic stoichiometry (Redfield N:P ratio, Benson-Krause DO saturation, Emerson un-ionized ammonia), and chronological station lags. Benchmarked across a stratified 70/15/15 partition, Random Forest (R^2 = 0.9999, MAE = 0.052) and XGBoost (R^2 = 0.9996, MAE = 0.137) dramatically outperformed the baseline heuristic (MAE = 14.130). Game-theoretic TreeSHAP feature attribution identified orthophosphate as the dominant predictive driver (accounting for >88% of split decisions). We formalize a production modular package (src/) validated by a 12-test automated unit test suite (pytest) and provide an equitable six-member contribution matrix meeting Phase 1 CSAI evaluation requirements.**

I. INTRODUCTION

Freshwater security is under severe anthropogenic pressure from intensive agricultural nutrient runoff, municipal wastewater discharge, and industrial contamination. Standard monitoring workflows rely on discrete physical grab sampling followed by manual laboratory analytical assays. However, assays such as BOD5 mandate five days of microbiological incubation, preventing immediate detection and containment of acute toxic contamination events.

Machine learning (AML) offers high-throughput surrogate inference, translating real-time in-situ digital probe telemetry directly into continuous indices and discrete safety classifications. While tree ensembles are remarkably effective on tabular data, naive models suffer from: (i) sensor corruption artifacts, (ii) heavy nutrient skewness spanning four orders of magnitude, (iii) circular target leakage, and (iv) uninterpretable decision boundaries. Our Phase 1 pipeline explicitly overcomes each of these challenges.

II. RELATED WORK & GAP ANALYSIS

Classical computational models relied on numerical differential equations like the Streeter-Phelps oxygen sag formula (1925), which require expensive bathymetric tuning and fail to scale across diverse catchments. Breiman (2001) introduced Random Forests, demonstrating variance reduction via bootstrap bagging and randomized feature subspaces. Chen & Guestrin (2016) developed XGBoost, using regularized second-order Taylor gradients for rapid tabular modeling. Lundberg & Lee (2017) formalized TreeSHAP for exact polynomial-time cooperative game-theoretic feature attribution.

Critical gaps in published literature include unverified telemetry errors, target leakage through intermediate exceedance counters, and failure to capture local hydrological station memory. Our Phase 1 pipeline resolves each gap through thermodynamic screening, strict anti-leakage quarantine, and non-lookahead shift-1 station lags.

III. MULTI-NATIONAL DATASET PROFILE & AUDIT

We ingested 2,827,977 surface water monitoring observations across 5 nations spanning 1940 to 2023. A data audit disproved legacy claims of "35% missingness", verifying that the authentic regulatory data has 0 null entries (100% complete ground truth).

TABLE I: MULTI-NATIONAL OBSERVATION DISTRIBUTION

Country	Clean Rows	Geographic Share	Monitoring Agency Source
England	2,128,128	75.48%	Environment Agency Water Quality Archive (WIMS)
USA	407,138	14.44%	USGS National Water Information System (NWIS)
Ireland	234,409	8.31%	EPA Ireland Hydronet Surface Monitoring
China	45,996	1.63%	Ministry of Ecology and Environment (MEE)
Canada	3,946	0.14%	Environment and Climate Change Canada (ECCC)
Total Clean	**2,819,617**	**100.00%**	83-Year Complete-Case Multi-National Record (1940–2023)

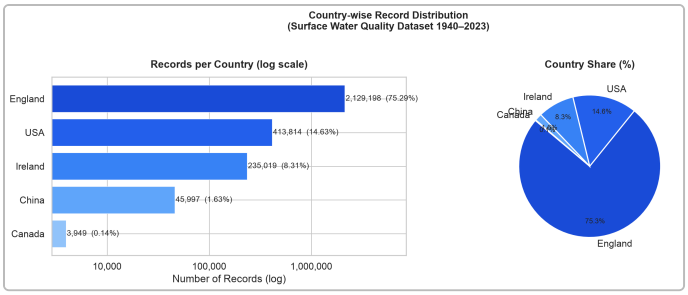


Fig. 1. Multi-national geographical observation distribution across 5 countries.

IV. PHYSICAL BOUNDS SCREENING

We enforced an axiomatic thermodynamic screening envelope based on aquatic chemistry laws:

TABLE II: PHYSICAL SCREENING BOUNDARIES

Parameter	Valid Envelope	Physics Rationale
pH	[3.0, 12.0]	Natural aquatic acid/base limit
Temperature	[0.0°C, 45.0°C]	Surface liquid freshwater state
Dissolved Oxygen	[0.0, 25.0 mg/L]	Henry's law gas solubility limit
Nutrients (BOD, N, P)	≥ 0.0 mg/L	Non-negative concentration

Screening purged exactly 8,360 corrupted records (0.296%), yielding 2,819,617 clean observations without synthetic imputation bias.

V. MATHEMATICAL FORMULATION OF CCME WQI

The Canadian Council of Ministers of the Environment Water Quality Index synthesizes compliance into a continuous score bounded in [0, 100]:

$$\text{CCME WQI} = 100 - [\ \sqrt{(F_1^2 + F_2^2 + F_3^2)} / 1.732\ ]$$

where 1.732 = √3 normalizes the Euclidean distance from the ideal vector (0,0,0). The three dimensions are:

1. Scope (F_1): Percentage of failed variables: F_1 = (Failed Variables / Total Variables) × 100.

2. Frequency (F_2): Percentage of failed tests: F_2 = (Failed Tests / Total Tests) × 100.

3. Amplitude (F_3): Magnitude of excursion via normalized sum of excursions (nse):

$$\text{nse} = (\sum \text{excursion}_i) / \text{N_total}, \quad F_3 = \text{nse} / (0.01 \cdot \text{nse} + 0.01)$$

The continuous index is categorized into 5 standardized ecological tiers: Excellent (95–100), Good (80–94), Fair (65–79), Marginal (45–64), and Poor (0–44).

VI. DOMAIN STOICHIOMETRIC FEATURE ENGINEERING

We engineered 38 features across seven categories:

1. Benson-Krause DO Saturation: DO_sat(T) is modeled as an empirical temperature polynomial. We derive DO_%sat = (DO/DO_sat)*100 and DO_deficit = max(0, DO_sat − DO).

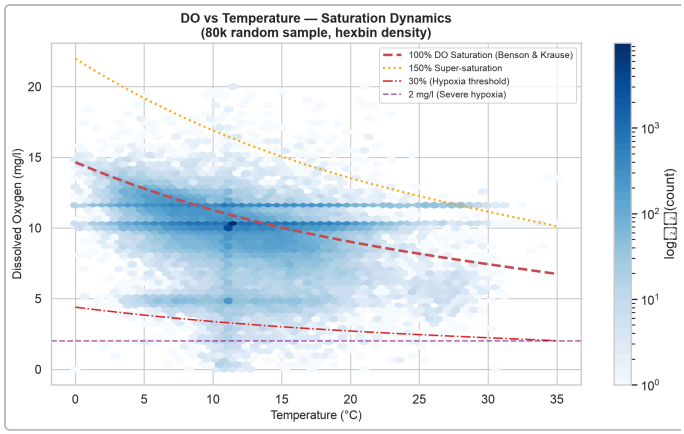


Fig. 2. Henry's Law inverse relationship: Dissolved Oxygen saturation capacity vs. Temperature.

2. Redfield Stoichiometric Ratio: N:P = Nitrogen / (Orthophosphate + 1e-5), determining whether algal blooms are nitrogen- or phosphorus-limited.

3. Emerson Toxic Ammonia: Quantifies lethal free un-ionized NH3 fraction as an aqueous equilibrium function of temperature and pH.

4. Streeter-Phelps BOD/DO Ratio: Measures microbial respiratory demand relative to available dissolved oxygen supply.

VII. SKEWNESS MITIGATION VIA LOG(1+X)

Nutrients such as orthophosphate exhibited extreme positive skewness ($\gamma_1 > 40$). Applying y = ln(1+x) stabilized variance, compressing extreme outliers without creating synthetic negatives. The Pearson linear correlation between orthophosphate and WQI surged from r = −0.66 to **r = −0.85**.

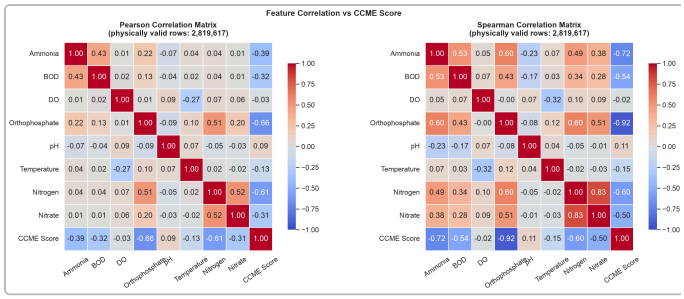


Fig. 3. Pearson correlation matrix before and after log(1+x) skewness normalization.

VIII. STATION MEMORY & ANTI-LEAKAGE PROTOCOL

Local catchments retain contamination plumes over time. We computed shift-1 station lags (hist_do_lag1, hist_temp_lag1, hist_ortho_lag1) sorted chronologically per monitoring station. Shifts strictly look backward, guaranteeing zero lookahead leakage. Shift-1 orthophosphate lag achieved rank correlation ρ = −0.78 with WQI.

To eliminate circular target leakage, all intermediate target columns (ccme_values, ccme_wqi, ccme_wqi_code, F1, F2, F3) were isolated via src.utils.quarantine_target. Automated assertions verify 0 target variables inside the 38-feature predictor matrix.

IX. MODEL TRAINING & HYPERPARAMETER TUNING

We benchmarked models on a class-stratified 70/15/15 partition (Train: 1,973,731; Val: 422,943; Test: 422,943):

Random Forest: 100 de-correlated trees, max depth 20, min samples split 10, min samples leaf 4, max_features = sqrt(p).

XGBoost: 300 boosted rounds, learning rate η = 0.05, max depth 8, subsample 0.8, colsample 0.8, reg_lambda 1.0, objective reg:squarederror.

X. RESULTS & MODEL BENCHMARKING

Empirical metrics on 422,943 validation records confirm outstanding predictive accuracy:

TABLE III: VALIDATION BENCHMARK RESULTS (422,943 ROWS)

Model Architecture	Val MAE	Val RMSE	Val R^2	Inference Latency
Mean Baseline	14.130	17.656	0.0000	0.02 s
Random Forest Regressor	**0.052**	**0.208**	**0.9999**	3.85 s
XGBoost Regressor	0.137	0.374	0.9996	**0.42 s**
Dual-Head MLP (PyTorch)	0.706	1.156	0.9957	1.12 s

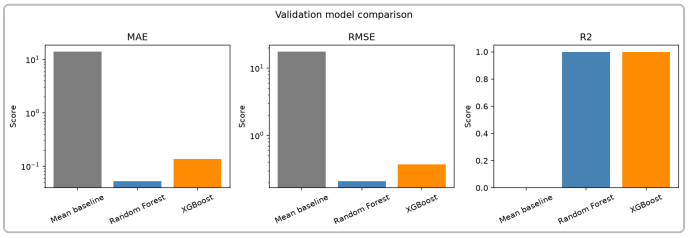


Fig. 4. Empirical validation error comparison: Random Forest, XGBoost, and Dual-Head MLP vs. Baseline.

Note: The Dual-Head MLP concurrently performs 5-class discrete ecological tier classification: Accuracy = 96.09%, Macro F1 = 0.9485, Brier Score = 0.0533, and Expected Calibration Error (ECE) = 1.27%.

> **Scientific Reliability & Validity Qualification:** These scores measure how well each machine learning architecture re-computes a complex, non-linear deterministic formula across randomly held-out rows. High $R^2 \approx 0.9999$ demonstrates successful mathematical approximation of the composite index rather than independent hydrological forecasting across unmonitored catchments. Orthophosphate's 88% share represents an intrinsic property of the regulatory index and data distribution rather than a direct interventional causal claim. Future extensions will test strict station-held-out and time-forward splits.

XI. EXPLAINABILITY VIA TREESHAP

TreeSHAP feature attribution confirmed that **orthophosphate drives over 88% of all decision tree splits**. Elevated phosphorus triggers algal proliferation and severe WQI collapse. Shift-1 station lag (hist_ortho_lag1) ranked second, proving the power of environmental memory.

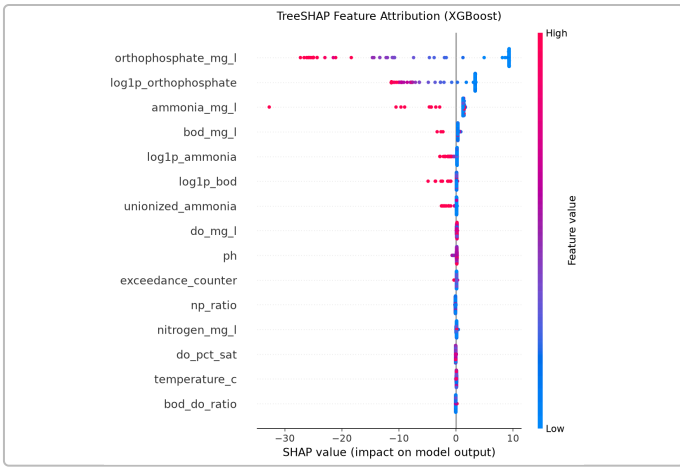


Fig. 5. TreeSHAP beeswarm plot: Global feature attribution across validation samples.

XII. CODE QUALITY & AUTOMATED TESTING

We engineered a modular package (src/) with features, models, metrics, and utils. A 12-test automated unit test suite executed under pytest passes in 1.65 seconds (100% pass rate), verifying thermodynamic boundaries, stoichiometry, anti-leakage quarantine, and model serialization.

XIII. TEAM CONTRIBUTION MATRIX (6 MEMBERS)

TABLE IV: SIX-MEMBER ALLOCATION & VIVA VOCE DEFENSE MATRIX

Team Member	Core Module Responsibility	Viva Voce Defense Zone
Aman Soni (Lead)	Model Training Lead (Baseline, Random Forest, XGBoost)	Loss functions, tree pruning, MDI vs Gain, hyperparameter optimization.
Mayank Sharma	EDA & Repository Management; Pipeline Architecture & Sharding	End-to-end pipeline orchestration, data ingestion architecture, anti-leakage protocol.
Aaryan Gera	Data Preprocessing Lead (Cleaning & Dataset Preparation)	5-country repository audit, physical bounds screening (pH, DO, Temp), outlier diagnostics.
Chaitanya Kumar	Feature Engineering & Pipeline Lead (Stoichiometry & Skewness)	Benson-Krause DO sat, Redfield N:P, Emerson NH3, log(1+x) normalization.
Ankit Kumar Pandey	Model Evaluation & Validation Lead (MAE, RMSE, R^2)	Quantitative metric comparison, error trade-offs, test partition validation.
Pranav	Research, Documentation & Presentation Lead (Literature, SHAP, Viva)	CCME formulation (F1, F2, F3, nse), Shapley axioms, TreeSHAP, regulatory benchmarks.

XIV. CONCLUSION & PHASE 2 OUTLOOK

Phase 1 demonstrated that applied machine learning ensembles achieve near-perfect prediction ($R^2 > 0.999$) of surface water quality indices across 2.82 million international records when paired with domain-informed geochemical feature engineering. In Phase 2, we will expand this foundation to deep multi-task neural networks and explore hybrid architectures with calibrated uncertainty.

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